

Forecasting Monthly Tea Production in Sri Lanka (2015-2025)

Group No: 05

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ABSTRACT

Tea production is a major contributor to Sri Lanka's economy, and understanding its temporal behaviour is essential for effective planning and decision-making. This study analyzes monthly tea production data from 2015 to 2025, focusing on total production, low elevation, and orthodox tea production.

Exploratory Data Analysis (EDA) revealed significant seasonal patterns, short-term fluctuations in production, and a noticeable structural decline in production after 2020. These findings highlight the dynamic nature of tea production and the need for appropriate forecasting techniques. To ensure stationarity, first order differencing was applied, and model identification was carried out using autocorrelation function (ACF) and partial autocorrelation function (PACF) analysis.

Based on model comparison criteria, including Akaike Information Criterion (AIC), Bayesian Information Criterion (BIC), and Ljung-Box test results, the SARIMA(1,1,2)(1,1,1,12) model was selected as the most suitable model for all three series. Diagnostic checks confirmed that the model residuals behave like white noise, indicating a good model fit.

Forecasts generated for the next six months suggest moderate fluctuations with a slight recovery trend in tea production, particularly during mid-year periods. The results demonstrate that time series models, particularly SARIMA, are effective tools for capturing seasonal behaviour and providing reliable short-term forecasts for tea production.

Keywords: Exploratory Data Analysis, Time Series Analysis, SARIMA, Tea Production, Forecasting

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1. INTRODUCTION

Tea production plays a vital role in Sri Lanka's agricultural and export sectors, contributing significantly to national income and providing employment opportunities to a large segment of the population. As one of the country's major export commodities, the tea industry is closely linked to economic stability and rural livelihoods. Therefore, accurate forecasting of tea production is essential for policymakers, plantation managers, and industry stakeholders to make informed decisions regarding production planning, resource allocation, and market strategies.

Tea production in Sri Lanka is influenced by several key determinants, among which elevation and processing methods play an important role. Elevation refers to the height of tea growing areas above mean sea level, measured in meters, and is classified into three categories: low grown (below 600 meters), medium grown (600-1200 meters), and high grown (above 1200 meters). These categories reflect variations in climatic conditions such as temperature and rainfall, which directly affect tea yield and quality.

In terms of processing methods, tea is broadly categorized into black tea and green tea. Black tea includes both orthodox tea and CTC tea, while green tea includes bio tea as a sub category. In this study, the analysis focuses specifically on elevationwise production and selected processing methods, particularly orthodox tea production, as key components influencing overall tea production patterns.

Time series analysis provides a systematic approach to examining data collected over time and identifying underlying patterns such as trends, seasonality, and fluctuations. These techniques are particularly useful in agricultural studies, where production levels are influenced by seasonal variations, climatic conditions, and other external factors. By analysing historical data, time series models can capture these patterns and generate reliable forecasts.

This study aims to analyse monthly tea production data from 2015 to 2025, focusing on total production, low elevation production, and orthodox tea production. The objective is to identify key patterns in the data and develop an appropriate forecasting model to generate short-term predictions, thereby supporting effective decision making in the tea industry.

2. LITERATURE REVIEW

Tea production is one of the most important sectors in the Sri Lankan economy, contributing significantly to export earnings, employment, and rural livelihood. The

industry supports a large proportion of the population and plays a key role in agricultural development. According to Kumarasinghe and Peiris (2018), tea cultivation contributes substantially to national income and provides employment opportunities to over one million people in Sri Lanka.

Due to its economic importance, accurate forecasting of tea production is essential for policymakers, producers, and stakeholders to make effective decisions regarding production planning and resource allocation.

Several studies have identified fluctuations and declining trends in tea production over time. Abeyanayake and Weerapura (2015) found that although tea production showed growth in earlier years, recent trends indicate a decline across all elevation categories, including high-grown, mid-grown, and low-grown tea.

Further evidence from the Tea Research Institute of Sri Lanka highlights that national tea production has been declining since 2013 due to factors such as reduced productivity, climate variability, and management issues. The report also shows that both yield and production levels have experienced downward trends in recent years.

These findings emphasize the need for reliable forecasting models to understand production behaviour and predict future trends accurately.

Time series analysis has been widely used in agricultural forecasting due to its ability to model temporal patterns such as trends and fluctuations. Among these methods, the Box-Jenkins ARIMA approach is one of the most commonly used techniques.

Kumarasinghe and Peiris (2018) applied ARIMA models to forecast annual tea production in Sri Lanka and identified different ARIMA structures for different elevation categories. Their study demonstrated that ARIMA models are effective in capturing production patterns and generating accurate forecasts.

Similarly, Abeynayake and Weerapura (2015) used both ARIMA and exponential smoothing techniques to analyse tea production trends. Their findings showed that these models can effectively represent time series data and provide reliable forecasts when evaluated using appropriate statistical criteria.

These studies confirm that classical time series models are suitable tools for forecasting tea production.

The theoretical foundation of time series forecasting is based on the Box-Jenkins methodology, which involves model identification, parameter estimation, and diagnostic checking. ARIMA models are particularly useful for handling non-stationary data through differencing techniques.

Hyndman and Athanasopoulos provide comprehensive guidelines for time series analysis, including concepts such as trend, seasonality, stationarity, and model evaluation. These concepts are essential for selecting appropriate models and ensuring the accuracy of forecasts.

Time series forecasting assumes that past patterns in the data will continue into the future, making it a valuable tool for predicting agricultural production.

Model evaluation is a crucial step in time series analysis to ensure the reliability of forecasts. Previous studies have used statistical measures such as Mean Absolute Percentage Error (MAPE) to assess forecasting accuracy.

Kumarasinghe and Peiris (2018) emphasized the importance of diagnostic checking, including the use of the Ljung-Box test and residual analysis, to verify whether the model assumptions are satisfied. Similarly, Abeynayake and Weerapura (2015) used residual diagnostics and MAPE to select the most appropriate models for tea production forecasting.

These evaluation methods help ensure that the selected model produces unbiased and accurate forecasts.

Although several studies have applied time series models such as ARIMA and exponential smoothing to forecast tea production, there are still some limitations. Most studies focus on annual data and do not explore detailed modeling using higher-frequency data.

Additionally, limited studies have conducted comprehensive comparisons of different time series models to identify the best performing model. There is also a need to validate models using separate datasets to ensure their predictive performance.

Therefore, this study aims to address these gaps by applying appropriate time series models and evaluating their performance to identify the most accurate model for forecasting tea production.

3. OBJECTIVES

The main objectives of this study are as follows:

- To analyse and forecast monthly tea production in Sri Lanka.
- To examine the pattern and characteristics of tea production data.
- To identify trends and seasonal variations in tea production.
- To develop an appropriate forecasting approach.
- To predict future tea production levels.

- To provide insights for decision-making.

4. DATA DESCRIPTION

The dataset consists of monthly tea production data in Sri Lanka from January 2015 to December 2025, totalling 132 observations. The variables considered in this study are:

- Total tea production
- Low elevation tea production
- Orthodox tea production

The data was structured as a time series with a monthly frequency. Any missing values were handled using interpolation techniques to ensure continuity.

5. EXPLORATORY DATA ANALYSIS (EDA)

5.1 TIME PLOT ANALYSIS

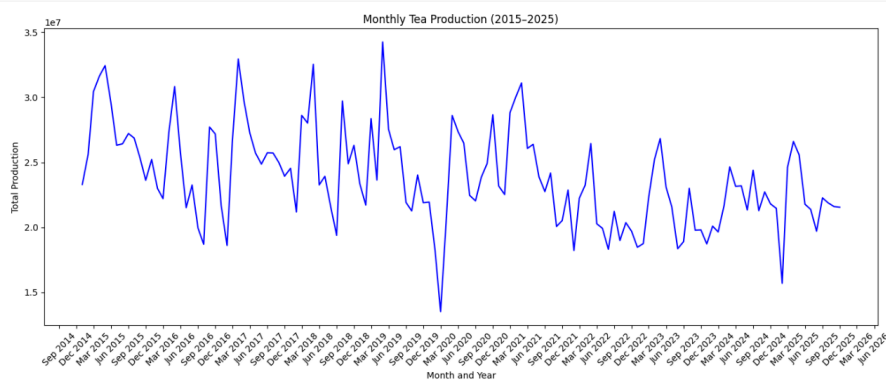


Figure 1: Monthly Tea Production (2015-2025)

As shown in Figure 1, the monthly time series plot of total tea production shows substantial short-term fluctuations across the study period. Production levels range approximately 15 million and 34 million units.

Although no strong long-term upward growth is observed, a noticeable decline appears after 2020, indicating possible structural changes in production dynamics. The series displays visible recurring peaks and troughs, suggesting the presence of seasonal behavior.

5.2 TREND ANALYSIS

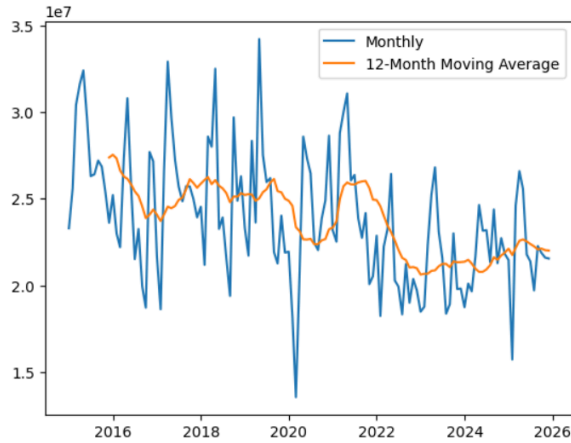


Figure 2: 12 Month Moving Average

As illustrated in Figure 2 a 12-month moving average was applied to smooth short-term fluctuations and highlighted the underlying trend in tea production. The smoothed trend reveals relatively stable and higher production levels during the period 2015-2019. A noticeable decline begins around 2020, followed by a more pronounced drop during 2022-2023. In recent years, a slight stabilization and partial recovery can be observed during 2024-2025. This pattern indicates the presence of non-stationary behaviour and suggests structural shifts in the production process.

5.3 SEASONAL ANALYSIS

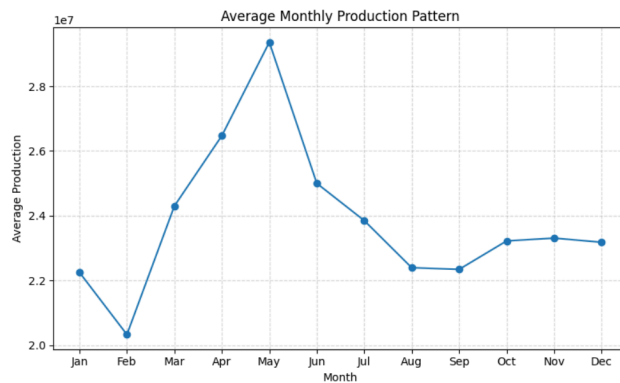


Figure 3: Average Monthly Production Pattern

As shown in Figure 3, the average monthly production pattern confirms the presence of strong seasonality in tea production. Production peaks around May, while the lowest production occurs around February. During the period from October to December, production levels remain relatively moderate. This repeating annual cycle indicates systematic seasonal variation, likely influenced by climatic conditions and harvesting

practices. The presence of such seasonality supports the use of seasonal time series models for analysis and forecasting.

5.4 ANNUAL PRODUCTION COMPARISON

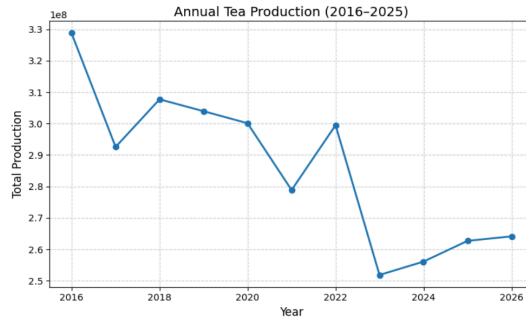


Figure 4: Annual Tea Production (2016-2025)

As illustrated in Figure 4, the annual tea production comparison highlights notable variations over time. The highest production was observed in 2016, reaching approximately 329 million units. Production levels remain relatively stable and high during the period from 2016 to 2019. However, a gradual decline begins in 2020, followed by a significant drop in 2023, where production reaches its lowest level at approximately 252 million units. A moderate recovery is observed during 2024-2025, although production levels do not return to those observed prior to 2020. This pattern suggests the presence of a structural break in tea production and indicates a shift in long-term production dynamics.

5.5 ELEVATIONWISE COMPARISON

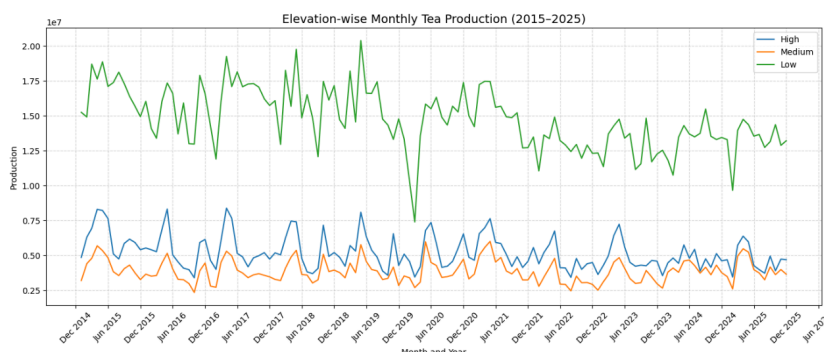


Figure 5: Elevationwise Monthly Tea Production (2015-2025)

As shown in Figure 5, tea production across elevation categories (high, medium, and low) reveals distinct contributions to total output. Low elevation areas contribute the highest

share of production, followed by medium elevation regions, while high elevation areas contribute comparatively less. Despite these differences in magnitude, all elevation categories exhibit similar seasonal patterns, indicating that climatic conditions and harvesting cycles affect all regions in a consistent manner.

5.6 PRODUCTION METHOD COMPARISON

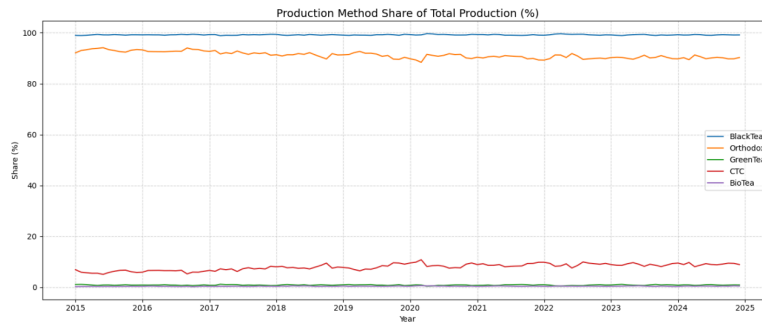


Figure 6: Comparison Between Production Methods

As illustrated in Figure 6, the analysis of tea production methods highlights the dominance of black tea in total production, contributing the largest shares throughout the study period. Within black tea production, orthodox tea maintains a consistently high contribution, indicating its importance in the industry. CTC tea contributes a smaller but noticeable share, while green tea and bio tea account for only a minimal proportion of total production. Overall, while total production fluctuates over time, the production structure remains largely stable. The dominance of black tea continues consistently, and no major structural shift in production methods is observed.

5.7 CROSS CORRELATION ANALYSIS

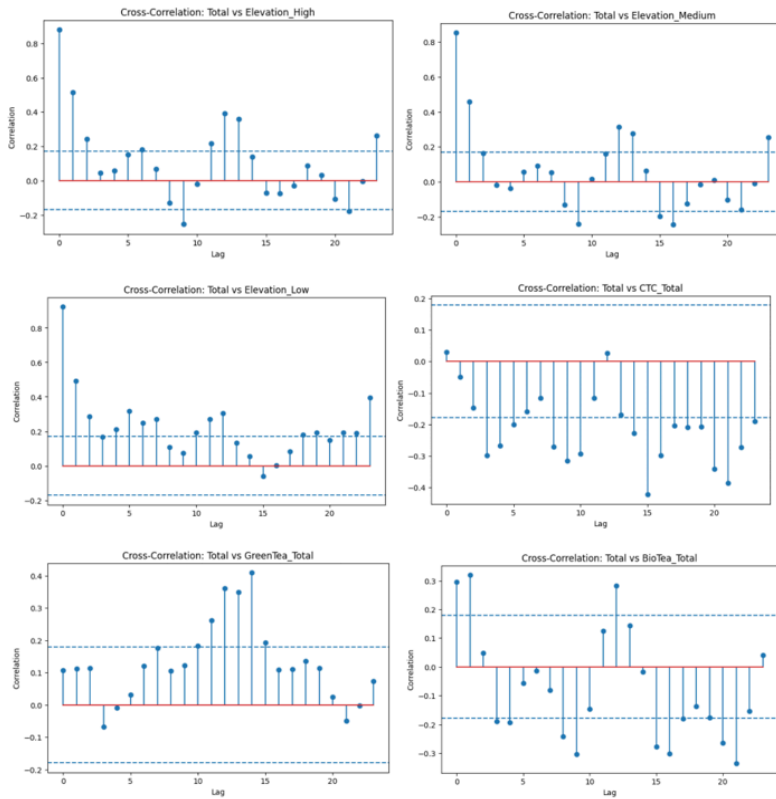


Figure 7: Cross Correlation

As shown in Figure 7, cross-correlation analysis was conducted between total production and selected elevation categories and production methods to examine potential lead-lag relationships.

The results show that for elevation categories, the strongest correlation occurs at lag 0. This indicates that changes in elevation-based production occur simultaneously with changes in total production. No consistent or significant lead-lag effects are observed beyond lag 0.

Similarly, for selected production methods, the strongest relationships are observed at lag 0. Although some minor correlations appear at other lags, they do not follow a consistent pattern.

Overall, the results indicate that production components move simultaneously with total production, and there is limited evidence of systematic delayed effects.

5.8 AUTOCORRELATION

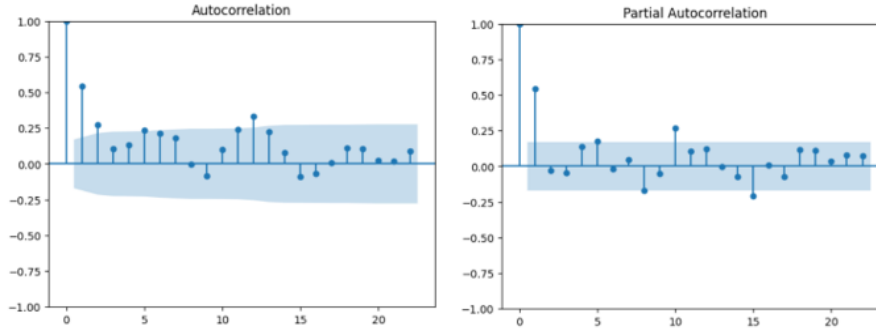


Figure 8: ACF and PACF Plot of Total Tea Production

As illustrated in Figure 8, the autocorrelation function (ACF) plot of total tea production shows significant positive autocorrelation at lower lags, along with a gradual decay pattern. This indicates strong temporal dependence in the data and suggests that the original series is non-stationary. Furthermore, the partial autocorrelation function (PACF) plot exhibits significant spikes at early lags, indicating the presence of autoregressive components in the series.

6. METHODOLOGY

6.1 STATIONARY

Stationary is a key requirement for time series modeling. The Augmented Dickey Fuller (ADF) test was used to assess stationarity. The results indicated that the original series were non-stationary due to the presence of trend and seasonality.

First order differencing was applied to remove non-stationarity. The differenced series exhibited stable mean and variance, confirming stationarity.

6.2 MODEL IDENTIFICATION

Several candidate models were estimated for each time series to identify the most appropriate model for forecasting. The selection of candidate models was based on the patterns observed in the ACF and PACF plots of the differenced series.

To evaluate model performance, statistical criteria such as the Akaike Information Criterion (AIC), Bayesian Information Criterion (BIC), and the Ljung-Box test were used. The AIC and BIC values measure the goodness of fit of the model while penalizing model complexity, with lower values indicating a better model. The Ljung Box test was used to assess whether the residuals are independently distributed, where a p-value greater than 0.05 indicated no significant autocorrelation.

Table 1 represents the comparison of candidate models for the time series.

Table 1: Model Comparison (Total Production, Low Elevation, Orthodox Tea)

Model	Variable	AIC	BIC	Ljung Box test
(1,1,2)(1,1,1,12)	Total Production	3414.873055	3430.739400	0.148521
(1,1,2)(1,1,1,12)	Low Elevation	3310.771499	3326.637845	0.146574
(1,1,2)(1,1,1,12)	Orthodox	3009.642081	3025.508426	0.176384
(0,1,1)(1,1,1,12)	Total Production	3447.116167	3457.732008	0.067868
(0,1,1)(1,1,1,12)	Low Elevation	3339.705488	3350.321330	0.250392
(0,1,1)(1,1,1,12)	Orthodox	3041.547085	3052.162926	0.082418
(1,1,1)(1,1,1,12)	Total Production	3448.140107	3461.409909	0.101636
(1,1,1)(1,1,1,12)	Low Elevation	3340.080526	3353.350327	0.100740
(1,1,1)(1,1,1,12)	Orthodox	3042.970692	3056.240494	0.110121
(1,1,1)(0,1,1,12)	Total Production	3449.148837	3465.072600	0.092916
(1,1,1)(0,1,1,12)	Low Elevation	3341.768032	3357.691794	0.162821
(1,1,1)(0,1,1,12)	Orthodox	3043.503732	3054.119574	0.092083
(2,1,1)(1,1,1,12)	Total Production	3449.505527	3460.121369	0.083456
(2,1,1)(1,1,1,12)	Low Elevation	3343.478500	3354.094341	0.142521
(2,1,1)(1,1,1,12)	Orthodox	3043.809301	3059.733064	0.115045
(1,1,0)(1,1,1,12)	Total Production	3493.910627	3504.564383	0.090696
(1,1,0)(1,1,1,12)	Low Elevation	3381.967681	3392.621438	0.220300
(1,1,0)(1,1,1,12)	Orthodox	3085.117721	3095.771477	0.112289

Among the candidate models, the *SARIMA*(1,1,2)(1,1,1,12) model consistently produced the lowest AIC and BIC values across all three series. Additionally, the Ljung Box test results for this model yielded p-value greater than 0.05, indicating that the residuals behave like white noise.

Although some alternative models showed comparable performance, they either had higher AIC values or did not perform as well in residual diagnostics. Therefore, the $SARIMA(1,1,2)(1,1,1,12)$ model was selected as the final model for all three series.

The consistency of the selected model across different variables further strengthens the robustness and reliability of the modelling approach.

6.3 MODEL ESTIMATION AND SELECTION

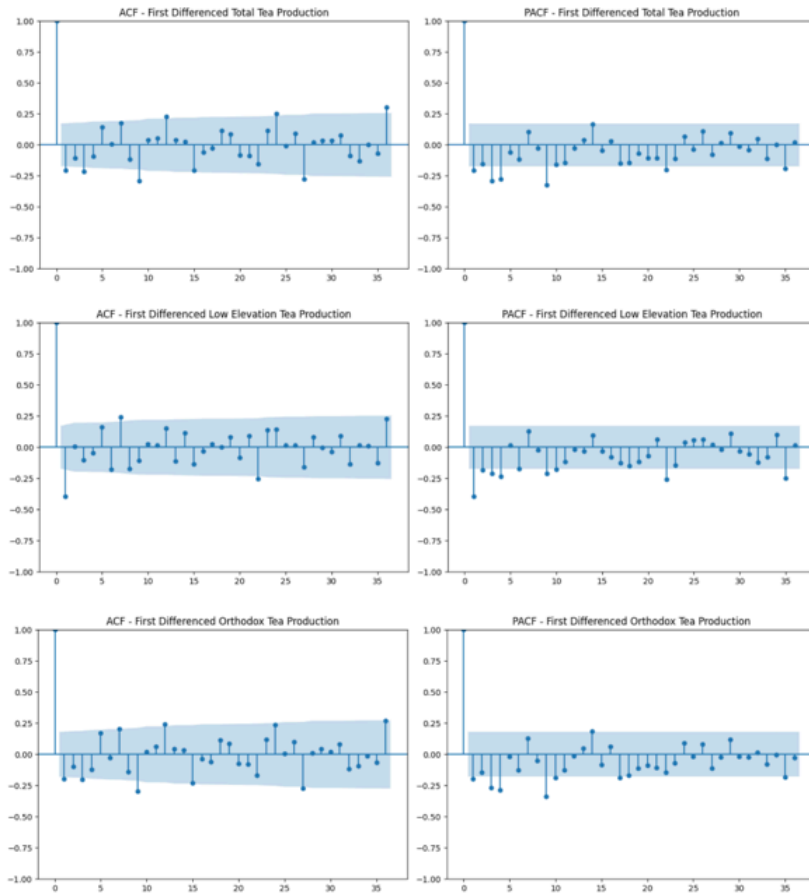


Figure 9: ACF and PACF of First Differenced Tea Production Series

To identify an appropriate model for forecasting tea production, several candidate time series models were estimated for each series, including total tea production, low elevation tea production, and orthodox tea production. The model identification process was guided by the patterns observed in the autocorrelation (ACF) and partial autocorrelation function (PACF) plots of the first-differenced series.

As shown in Figure 9, the ACF plots exhibit a gradual decay pattern, while the PACF plots display significant spikes at lower lags. This behavior suggests the presence of both autoregressive (AR) and moving average (MA) components in the data. Additionally, the

presence of noticeable spikes at seasonal lags indicates strong seasonal behaviour, supporting the inclusion of seasonal components in the model. Based on these observations, several candidate Seasonal Autoregressive Integrated Moving Average (*SARIMA*) models were considered for further evaluation.

To select the most suitable model, statistical criteria including the Akaike Information Criterion (AIC), Bayesian Information Criterion (BIC), and the Ljung-Box test were employed. The AIC and BIC values measure the goodness of fit of the model while penalizing model complexity, with lower values indicating better model performance. The Ljung-Box test was used to examine whether the residuals are independently distributed. A p-value greater than 0.05 indicates that the residuals behave like white noise, suggesting an adequate model.

Table 1 presents the comparison of candidate models for each series based on these criteria.

Among the evaluated models, the *SARIMA*(1,1,2)(1,1,1,12) model consistently produced lowest AIC and BIC values across all three series. Furthermore, the Ljung-Box test results for this model yielded p-values greater than 0.05 for all series (Total Production: 0.148, Low Elevation: 0.147, Orthodox Tea: 0.176), indicating that the residuals do not exhibit significant autocorrelation.

Although alternative models showed reasonable close performance, they either had higher AIC/BIC values or did not perform as well in diagnostic checks. Therefore, the *SARIMA* (1,1,2)(1,1,1,12) model was selected as the final model for all three production series.

The consistency of the selected model across different categories further strengthens the reliability and robustness of the modeling approach, making it suitable for forecasting tea production in Sri Lanka.

6.4 MODEL ESTIMATION RESULTS

Following the model selection process described in Section 7.3, the *SARIMA* (1,1,2)(1,1,1,12) model was selected as the most appropriate model for all three production series. The model was then fitted to each series to estimate the parameters and evaluate its overall performance.

The estimation results indicate that both non-seasonal and seasonal components play an important role in explaining the behaviour of tea production. The inclusion of first-order differencing ($d = 1$) effectively removes non-stationarity in the series, while seasonal

differencing ($D = 1, s = 12$) captures the annual seasonal patterns observed in tea production.

The autoregressive (AR) and moving average (MA) components account for short term dependencies and random fluctuations in the data. In particular, the inclusion of two moving average terms and one autoregressive term allows the model to effectively capture the dynamic structure of the series.

The summary of the estimated models for each series is represented in Table 2.

Table 2: Summary of Estimated SARIMA Models

Series	AR Terms	MA Terms	Seasonal Terms	AIC	BIC
Total Production	AR(1)	MA(2)	(1,1,1,12)	3414.87	3430.74
Low Elevation	AR(1)	MA(2)	(1,1,1,12)	3310.77	3326.64
Orthodox Tea	AR(1)	MA(2)	(1,1,1,12)	3009.64	3025.51

As shown in Table 2, the selected $SARIMA(1,1,2)(1,1,1,12)$ model provides a good fit for all three series, as indicated by relatively low AIC and BIC values. Among the three series, orthodox tea production records the lowest AIC and BIC values, suggesting a slightly better fit compared to the other series.

The results confirm that the inclusion of both non-seasonal and seasonal components is essential for accurately capturing the underlying structure of tea production data. Although some individual parameters may not be statistically significant, the overall model performance remains strong, as evidenced by the model selection criteria and subsequent diagnostic checks.

Overall, the estimation results demonstrate that the selected $SARIMA$ model effectively captures both short-term fluctuations and seasonal patterns in tea production, providing a reliable foundation for diagnostic checking and forecasting.

6.5 DIAGNOSTIC CHECKING

Following the estimation of the $SARIMA(1,1,2)(1,1,1,12)$ models, diagnostic tests were conducted to assess the adequacy and reliability of the fitted models for total tea production, low elevation tea production, and orthodox tea production.

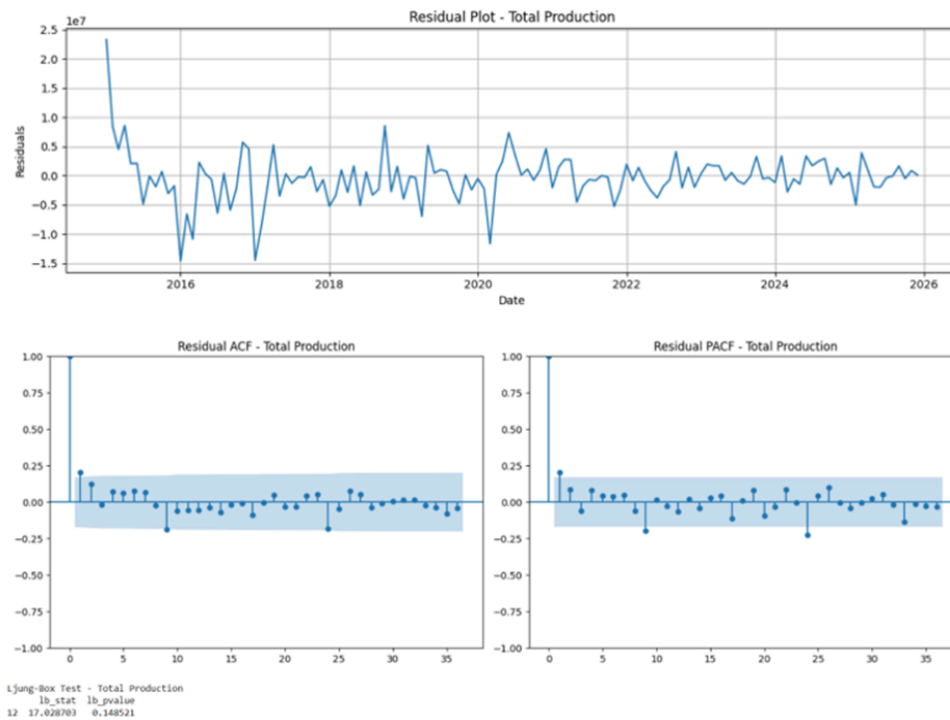


Figure 10: Residual Diagnostic of SARIMA Model - Total Production

The residual diagnostics for total tea production are presented in Figure 10. As shown in Figure 10, the residuals fluctuate randomly around zero without any clear trend or systematic pattern, indicating that the model has successfully captured the underlying structure of the data. Furthermore, the ACF and PACF plots of the residuals show that most of the autocorrelation values lie within the confidence bounds, suggesting the absence of significant autocorrelation.

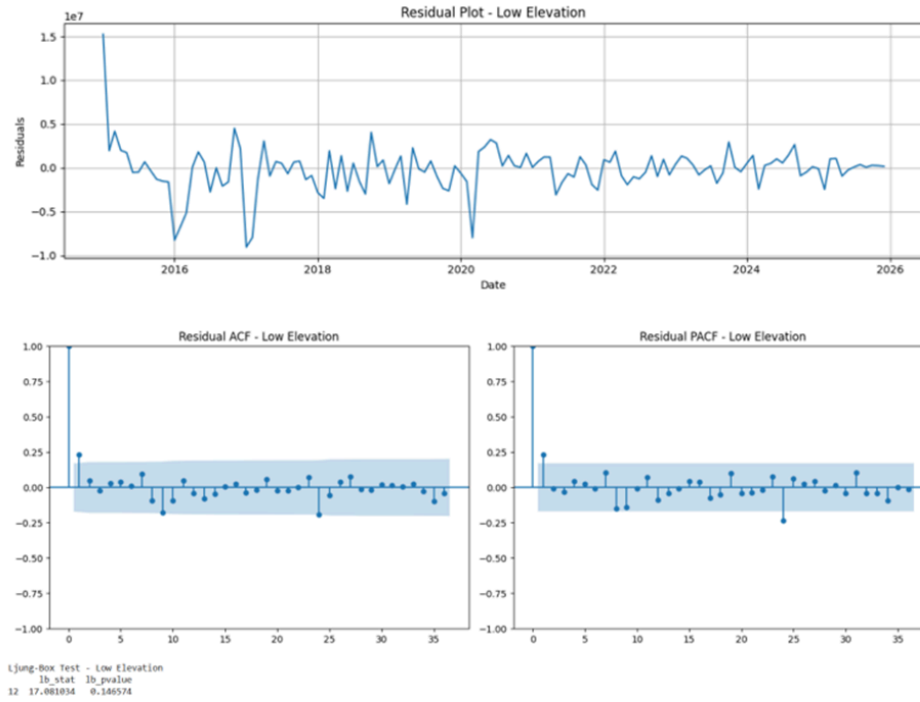


Figure 11: Residual Diagnostic of SARIMA Model - Low Elevation

The residual diagnostics for low elevation tea production are presented in Figure 11. Similar to the total production series, the residuals exhibit random variation around zero, and the ACF and PACF plots indicate no remaining autocorrelation. This confirms that the model adequately captures the temporal dynamics of the low elevation series.

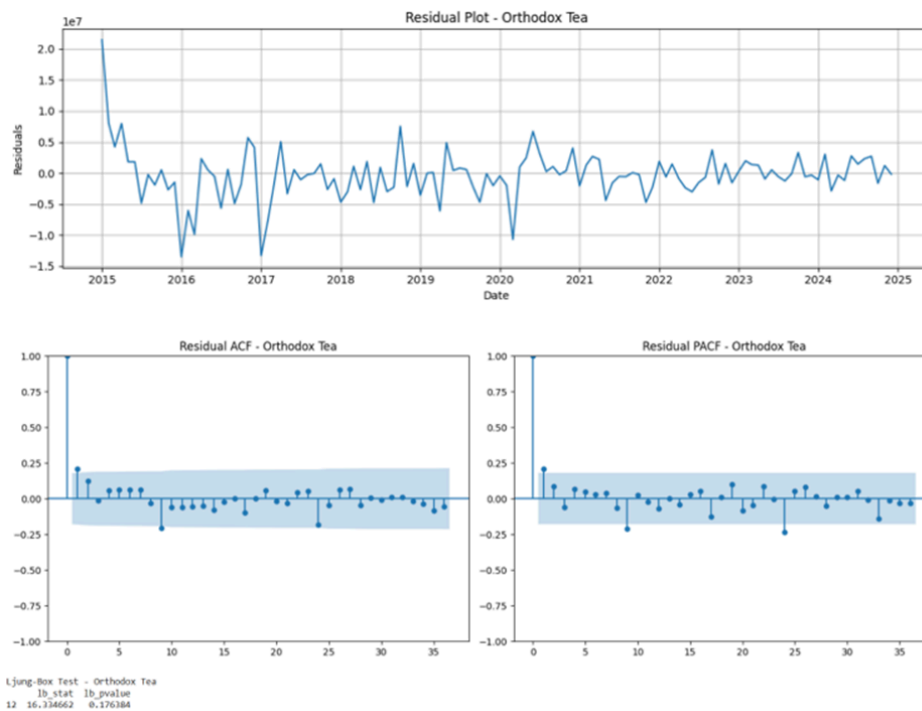


Figure 12: Residual Diagnostic of SARIMA Model - Orthodox Tea

The residual diagnostics for orthodox tea production are shown in Figure 12. The residuals again display no systematic pattern, and the corresponding ACF and PACF plots suggest that the residuals behave like white noise. This indicates that the model provides a satisfactory fit for this series as well.

In addition to graphical diagnostics, the Ljung-Box test was conducted to statistically verify the independence of residuals. The test results yield p-values greater than 0.05 for all three series (Total Production: 0.148, Low Elevation: 0.147, Orthodox Tea: 0.176), indicating that the null hypothesis of no autocorrelation cannot be rejected. This confirms that the residuals are independently distributed.

Although minor deviations from normality are observed, such as those indicated by the Jarque-Bera test, these are common in practical time series applications and do not significantly affect the forecasting performance of the models.

Overall, both graphical and statistical diagnostic results confirm that the selected $SARIMA(1,1,2)(1,1,1,12)$ models are statistically adequate and suitable for forecasting tea production.

6.6 FORECASTING RESULTS

Using the selected $SARIMA(1,1,2)(1,1,1,12)$ model, forecasts were generated for the next six months for the total tea production, low elevation tea production, and orthodox tea production. The forecast results, along with their confidence intervals, are illustrated in Figures 13-15.

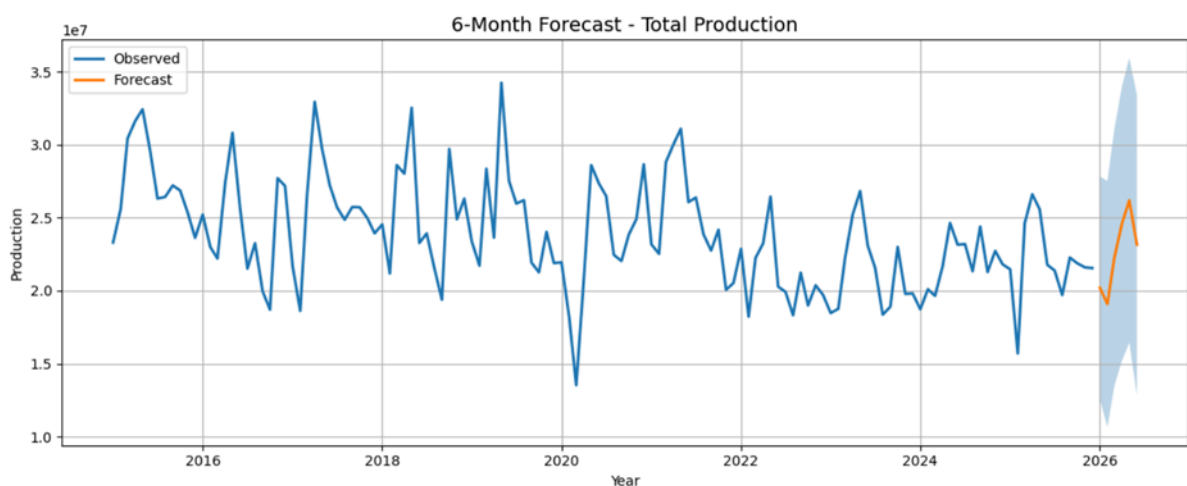


Figure 13: 6-Month Forecast - Total Tea Production

The six month forecast for total tea production is presented in Figure 13. The results indicate a moderate upward trend over the forecast horizon. Production is expected to

increase from approximately 20.2 million units in January 2026 to around 26.2 million units in May 2026, followed by a slight decline in June 2026. This pattern reflects the seasonal fluctuations observed in the historical data.

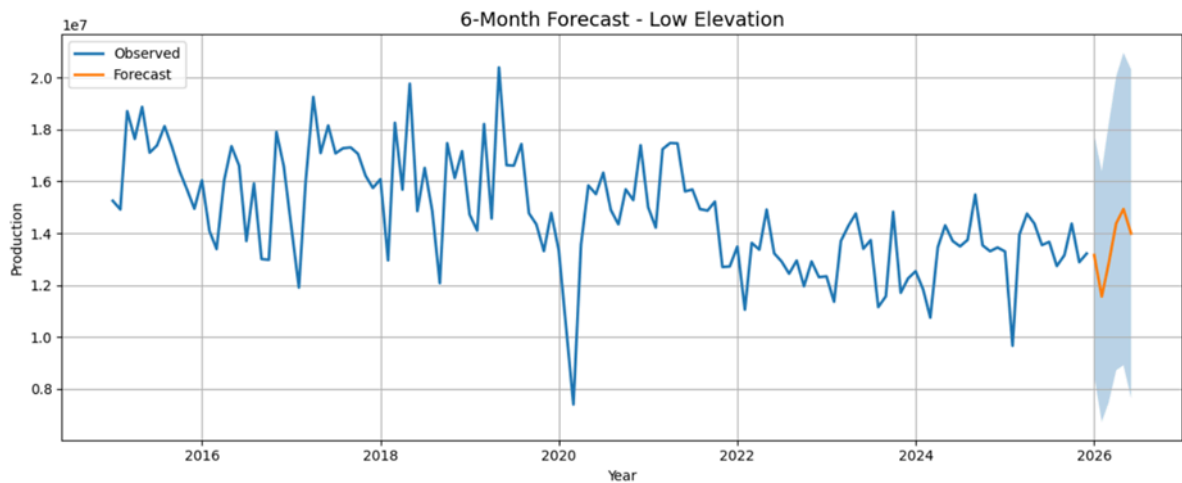


Figure 14: 6-Month Forecast - Low Elevation

The forecast for low elevation tea production is shown in Figure 14. As the largest contributor to total production, the low elevation series is expected to fluctuate between approximately 11.5 million and 14.9 million units during the forecast period. Despite short-term variations, the overall pattern remains relatively stable, reinforcing its dominant role in tea production.

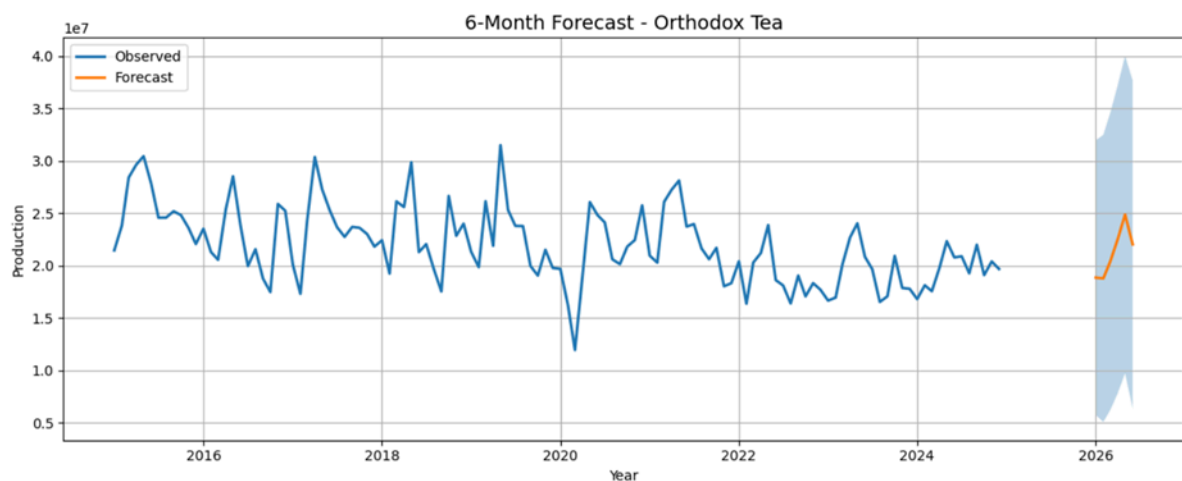


Figure 15: 6-Month Forecast - Orthodox Tea

The forecast results for orthodox tea production are presented in Figure 15. The series is projected to exhibit a gradual increase during the forecast period, reaching a peak around May 2026 before slightly declining in June 2026. This behaviour is consistent with the seasonal patterns identified in the earlier analysis.

In all forecast plots, the confidence intervals widen as the forecast horizon increases, reflecting the growing uncertainty associated with longer term predictions. This is a common characteristic of time series forecasting models.

Overall, the forecasting results suggest a modest recovery in tea production in the near future. However, production levels remain below the peaks observed in earlier years, indicating that the sector has not yet fully returned to its previous performance levels. These findings provide useful insights for policymakers and industry stakeholders in planning and decision making related to tea production.

7. CONCLUSION

This study examined the temporal behaviour of tea production in Sri Lanka using monthly data from 2015 to 2025, focusing on total production, low elevation production, and orthodox tea production. The analysis revealed that tea production exhibits strong seasonal patterns, noticeable short term fluctuations, and a structural decline after 2020, indicating changes in production dynamics over time.

Exploratory Data Analysis provided important insights into trends, seasonality, and variability in the data. The presence of non-stationarity was addressed through first order differencing, which enabled the application of time series modeling techniques. Based on the analysis of autocorrelation and partial autocorrelation functions, several candidate models were evaluated.

Among the models considered, the SARIMA(1,1,2)(1,1,1,12) model was identified as the most suitable of all three series based on AIC, BIC, and Ljung-Box test results. Diagnostic checking confirmed that the residuals behave like white noise, indicating that the model adequately captures the underlying structure of the data.

The forecasting results suggest moderate fluctuations with a slight recovery trend in tea production in the near future. However, production levels remain below previous peaks, highlighting the need for strategic interventions.

Overall, this study demonstrates that time series models are effective tools for analyzing and forecasting tea production, providing valuable insights for policymakers and industry stakeholders.

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